

HARRISON HAYWARD

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EDUCATION

Carnegie Mellon University | College of Engineering

Expected Graduation: May 2027

M.S. in Electrical and Computer Engineering | Minor in Human-Computer Interaction

Pittsburgh, PA

Relevant Coursework: Electronic Devices and Analog Circuits, Natural Language Processing, Structure and Design of Digital Systems, Fundamentals of Semiconductor Devices, Microelectronic Circuits, Introduction to Computer Architecture

PROJECTS

Pipelined RISC-V CPU

March 2026

- Designed a 7-stage RISC-V processor in SystemVerilog with hazard detection, forwarding, and a 2-bit branch predictor.
- Reduced pipeline stalls to zero, improving IPC from 0.38 to 0.99 leading to a 2.5x higher throughput via forwarding.
- Achieved 1.72x wall-clock performance speedup and a 27.5% cycle reduction using branch prediction.

Mixed Signal IC Design

December 2025

- Simulated an 8-bit R-2R DAC and CMOS OTA buffer using Cadence Virtuoso.
- Analyzed output impedance, reducing load-induced voltage drop from 35% to <0.1% using an OTA buffer.
- Designed semiconductor layout under strict area constraints utilizing chaining, shared wells, and capacitor integration.

Hardware Thread Design

March 2025

- Designed Bluetooth transmitter and receiver hardware threads in SystemVerilog.
- Developed and ran the data through a SECDED (Single Error Correction, Double Error Detection) decoder.
- Tested and deployed these hardware threads on FPGA.

NLP Design and Training

March 2025

- Created, trained, and tested a transformer-based large language model.
- Honed model hyperparameters and applied prompt engineering to improve accuracy and F1 score.
- Implemented models with PyTorch using local GPU acceleration to increase training efficiency.

EXPERIENCE

HPS Investment Partners (BlackRock)

June 2025 - August 2025

Data Analyst Intern

New York, NY

- Developed an SQL stored procedure to derive tranches of investment securities.
- Employed query optimization strategies to improve efficiency on large datasets.
- Built a Power BI dashboard to calculate LTVs of investments enabling faster risk assessment.
- Delivered live presentation of final project to entire Data Analytics team and management with active Q&A session.

GoldenTree Asset Management

June 2024 - August 2024

Data Analyst Intern

New York, NY

- Constructed Excel models and utilized Bloomberg data sources to forecast company financial health and performance.
- Employed Google BigQuery, and Tableau to create dashboards of large datasets for research analysts.
- Applied SQL to compare fund-level performance metrics with ICE benchmarks.

NYC FIRST

June 2021 - August 2022

Web Design Intern

Cornell Tech, New York, NY

- Designed and launched a makerspace program for underserved NYC students, bridging access gaps.
- Constructed an online course platform to support remote learning and reduce in-person instruction barriers.
- Applied Fusion 360 and data analysis to adapt robotics education and designed COVID-safe gear during the pandemic.

LEADERSHIP AND AWARDS

Track and Field Team - Hackley High School and CMU Division III - Pole Vault

September 2017 - December 2025

- Pole Vaulted with Men's Track and Field team; Personal Record: 14'6".
- Men's Outdoor UAA Champions (2023 & 2024).
- NYSAIS Pole Vault Outdoor Champion (May 2022).
- Competed in NYPHSAA (NY State) Indoor and Outdoor Pole Vault Championships (March - May 2022).
- Qualified for New Balance Indoor Pole Vault National Championship (March 2022).

CMU Orientation Leader

August 2025

- Led 3 Orientation Counselors and 40 incoming students achieving nearly 100% attendance throughout the 9 days.
- Planned and set up events involving all 1800 students. Involved delegation, emergency response, and teamwork.

SKILLS

Languages: C, C++, Python, SQL, HTML/CSS, Java, JavaScript

Hardware: SystemVerilog, Cadence Virtuoso, Vivado, Fusion 360, Jetson

Professional: Mentorship and Coaching, Rapid Iteration, Execution Under Deadlines, Project Ownership